

Drive Cokoino 4WD Robot Hat Shield Shield to run the chassis

Table

1. Preface	2
2. Component List	2
3. Component related knowledge	3
3.1 Knowledge of the TT Motor	3
3.2 Knowledge of Cokoino 4WD Robot Hat Shield	4
3.2.1 overview	4
3.2.2 Appearance of the Cokoino 4WD Robot Hat Shield	5
3.2.3 Introduction for the Cokoino 4WD Robot Hat Shield	6
3.2.4 DRV8833 Specification	8
3.2.5 Safety and notes	9
3.2.6 Power supply	9
3.2.7 Pin Description of the 4WD Robot Hat	10
3.2.8 DC motor interfaces on the 4WD Robot Hat	11
3.3 Knowledge of the Raspberry Pi 4B	13
4. Circuit Connection	16
4.1 Circuit connection diagram:	18
5. Download Code and Run	19
6. Trouble shooting	23
7. Any questions and suggestions are welcome	24

1. Preface

Our final form of this product is a small car chassis with one servo, 2 motors and 4 wheels, without motor drive modules, control boards, batteries, and other things. Its highlight lies in its development and scalability. You can choose the motor drive and control board you want to use, install it on the chassis of this car, and make it run, becoming a four-wheel drive car. This will be a challenging and fulfilling task. Wishing you success!

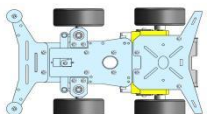
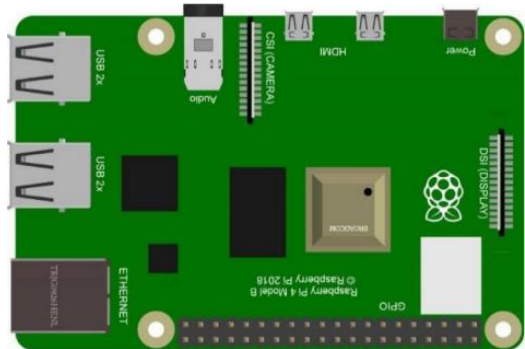
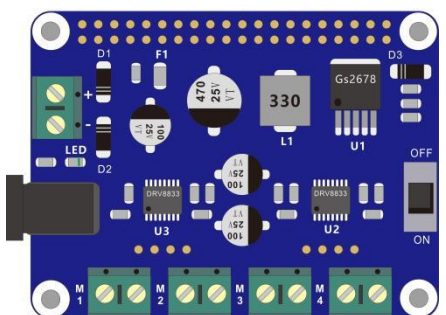
You can also refer to our Demo2, where we assembled Cokoino 4WD Robot Hat Shield , Raspberry Pi 4B, and 2pcs 18650 batteries onto the chassis of the car, creating a car based on Raspberry Pi. The following tutorial will provide a detailed introduction to Demo2, including component list, component related knowledge, circuit connection, code, and more. If you are interested in demo2, you can refer to the demo1 checklist to prepare relevant materials for experimentation.

If you have any technical issues or suggestions, please provide feedback to us via email:

cokoino@outlook.com

2. Component List

For this Demo experiment, what do you need to prepare like below list

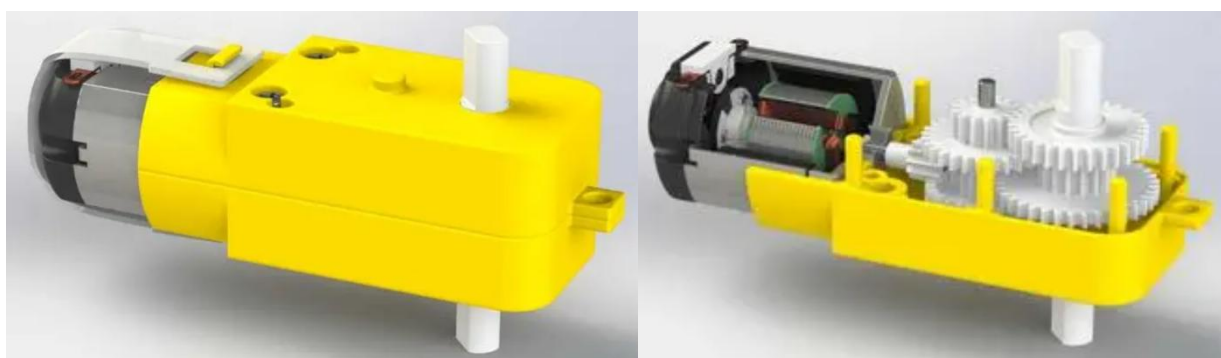
Component/Module	QTY	Picture	Remark
Real-Wheel Drive Steering Car Chassis	1		Provided by the Real-Wheel Drive Steering Car Chassis Kit,you need assembled by yourself
Raspberry Pi 4B	1		Not included in this Kit You need to prepare these by yourself.
Cokoino 4WD Robot Hat Shield	1		

18650 battery	2		Not included in this Kit You need to prepare these by yourself.
Male to Female dupont wire	3		

3. Component related knowledge

3.1 Knowledge of the TT Motor

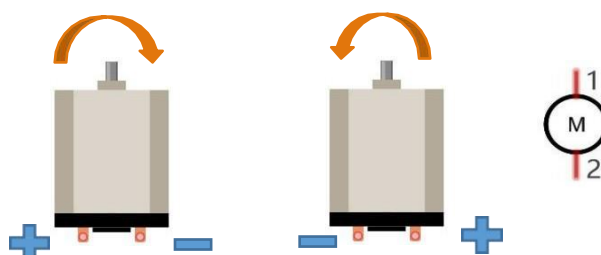
As shown in the figure below, the TT motor consists of a DC motor and related gears, with a yellow outer shell fastened.



DC Motor

When motor is connected to the power supply, it will rotate in one direction. Reverse the polarity of power supply, the motor will rotate in the opposite direction.

And the speed of motor depends on the voltage between two ends. The larger the voltage, the larger the speed.

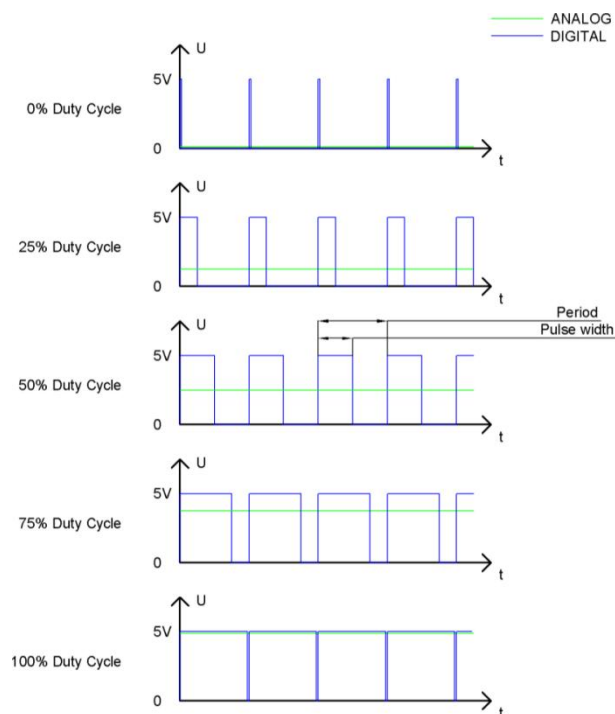


PWM

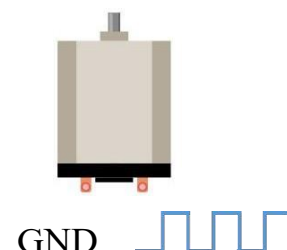
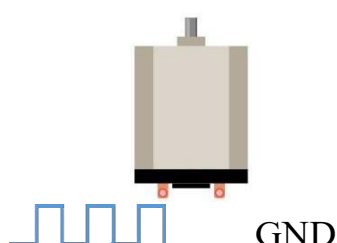
PWM, Pulse Width Modulation, uses digital pins to send certain frequencies of square

waves, that is, the output of high levels and low levels, which alternately last for a while. The total time for each set of high levels and low levels is generally fixed, which is called the period (the reciprocal of the period is frequency). The time of high level outputs are generally called “pulse width”, and the duty cycle is the percentage of the ratio of pulse duration, or pulse width (PW) to the total period (T) of the waveform.

The longer the output of high levels last, the larger the duty cycle and the higher the corresponding voltage in analog signal will be. The following figures show how the analog signal voltage vary between 0V-5V (high level is 5V) corresponding to the pulse width 0%-100%:



The longer the PWM duty cycle is, the higher the output power will be. Now that we understand this relationship, we can use PWM to control the brightness of an LED or the speed of DC motor and so on.

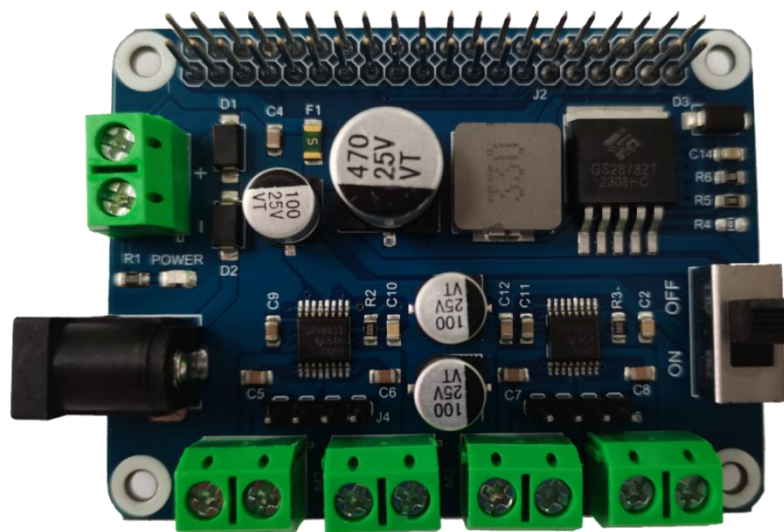


3.2 Knowledge of Cokoino 4WD Robot Hat Shield

3.2.1 overview

Cokoino 4WD Robot Hat Shield is a Raspberry Pi power and motor driver board suitable for Raspberry

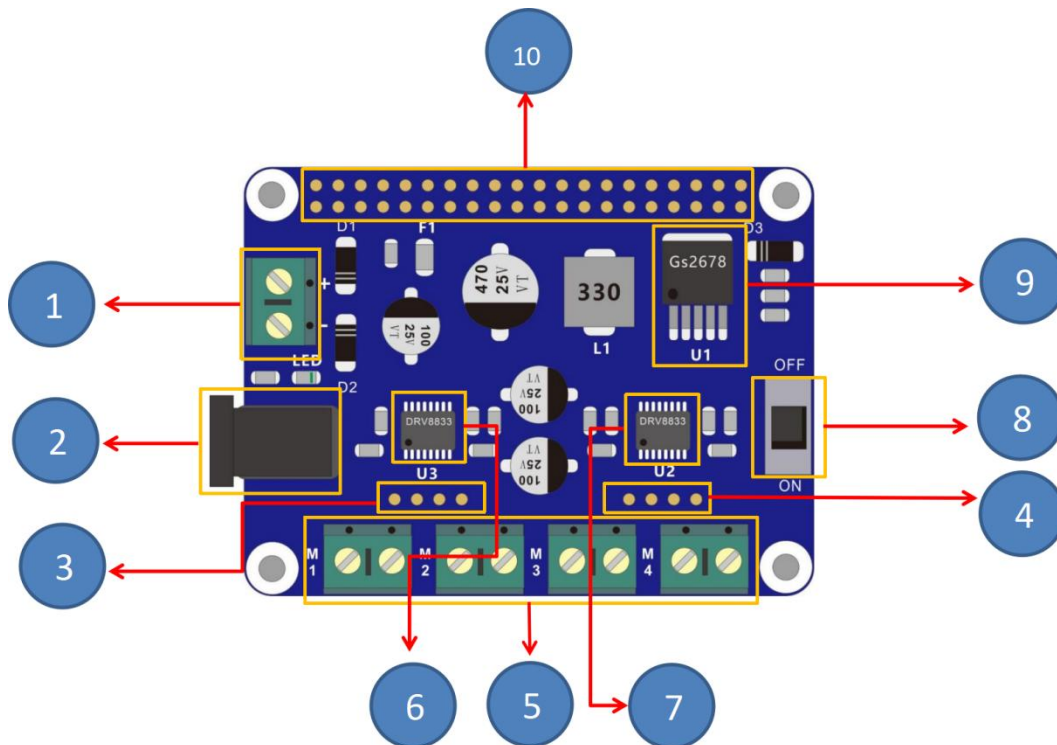
It has already been listed on Cokoino's Amazon store. If you like it, you can purchase it from Cokoino's Amazon store.



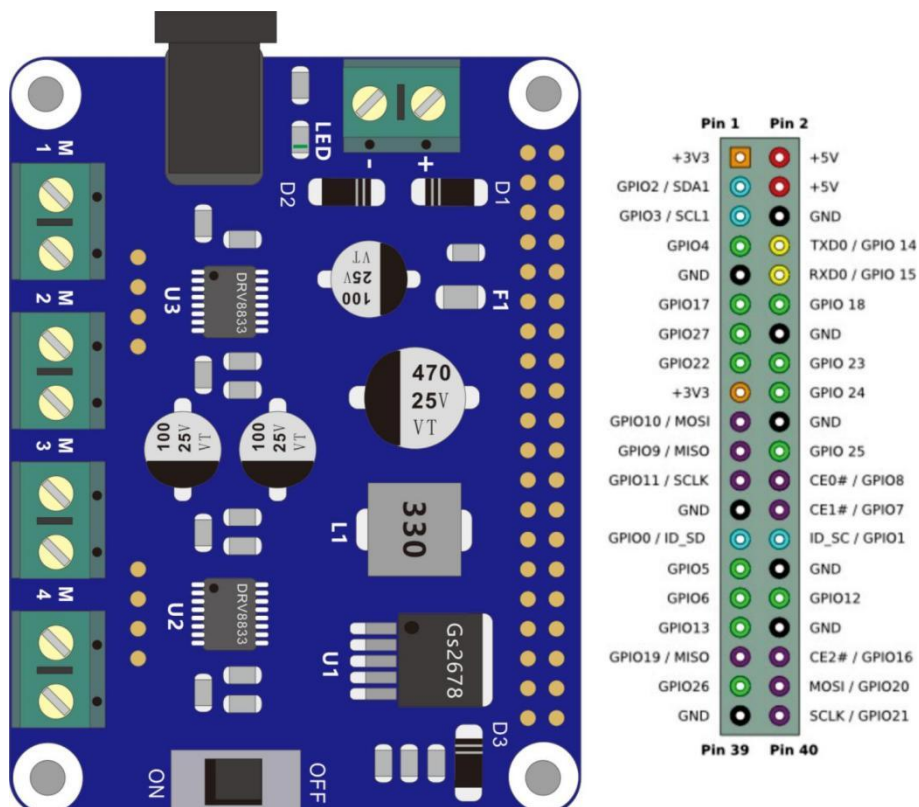
3.2.3 Introduction for the Cokoino 4WD Robot Hat Shield

The Cokoino 4WD Robot Hat Shield is suitable for Raspberry Pi 4B/3B/3B+/3A+/2B/1B+/1A+/Pi Zero/Pi Zero W, using a plastic long needle double row female. The row female corresponds to the insertion of Raspberry Pi, and the long needle corresponds to the signal of each pin of Raspberry Pi.

For the main components and functions of the Cokoino 4WD Robot Hat Shield, please refer to the following table.



Pin distribution of Pi Power&4WD HAT relative to Raspberry Pi



No.	Component	Function
1	Power port(terminal)	Power input 2pin port, pay attention to the positive and negative when connecting the external power supply, power supply range: 7-12V DC
2	Power port(DC port)	Power input DC port, power supply range: 7-12V DC
3	Motor interface (row pin)	2.54 * 4 pin row pin, corresponding to circuit M1, M2 motor interface
4	Motor interface (row pin)	2.54 * 4 pin row pin, corresponding to circuit M3, M4 motor interface
5	Motor interface (terminal)	KF301 two pin terminals, corresponding to circuit M1, M2, M3, M4 motor interface
6	DRV8833	It is motor driver chip that can drive two DC Motors
7	DRV8833	It is motor driver chip that can drive two DC Motors
8	Power Switch	Dial to ON to power on, dial to OFF to power off
9	GS2678	GS2678 is an asynchronous DC buck converter with PWM voltage management IC and pulse width modulation output (PWM). The input voltage range can be from a minimum of 3.6V to a maximum of 32V, the output voltage can be adjusted from 0.8V-23V, and the output current can reach up to 5A
10	RaspberryPi interface	Plastic long needle double row female, the row female is connected to the row pin interface of the Raspberry Pi, and the long needle is used to expand the pins of the Raspberry Pi to the top of the Cokoino 4WD Robot Hat Shield

3.2.4 DRV8833 Specification

DRV8833 Dual H-Bridge Motor Driver

1 Features

- Dual-H-Bridge Current-Control Motor Driver
 - Can Drive Two DC Motors or One Stepper Motor
 - Low MOSFET ON-Resistance: HS + LS 360 mΩ
- Output Current (at $V_M = 5\text{ V}$, 25°C)
 - 1.5-A RMS, 2-A Peak per H-Bridge in PWP and RTY Package Options
 - 500-mA RMS, 2-A Peak per H-Bridge in PW Package Option
- Outputs can be in Parallel for
 - 3-A RMS, 4-A Peak (PWP and RTY)
 - 1-A RMS, 4-A Peak (PW)
- Wide Power Supply Voltage Range: 2.7 to 10.8 V
- PWM Winding Current Regulation and Current Limiting
- Thermally Enhanced Surface-Mount Packages

2 Applications

- Battery-Powered Toys
- POS Printers
- Video Security Cameras
- Office Automation Machines
- Gaming Machines
- Robotics

3 Description

The DRV8833 device provides a dual bridge motor driver solution for toys, printers, and other mechatronic applications.

The device has two H-bridge drivers, and can drive two DC brush motors, a bipolar stepper motor, solenoids, or other inductive loads.

The output driver block of each H-bridge consists of N-channel power MOSFETs configured as an H-bridge to drive the motor windings. Each H-bridge includes circuitry to regulate or limit the winding current.

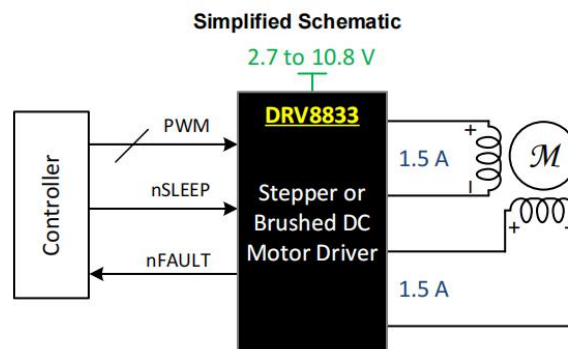
Internal shutdown functions with a fault output pin are provided for overcurrent protection, short-circuit protection, undervoltage lockout, and overtemperature. A low-power sleep mode is also provided.

The DRV8833 is packaged in a 16-pin WQFN package with PowerPAD™ (Eco-friendly: RoHS & no Sb/Br).

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
DRV8833	TSSOP (16)	5.00 mm × 4.40 mm
	HTSSOP (16)	5.00 mm × 4.40 mm
	WQFN (16)	4.00 mm × 4.00 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



3.2.5 Safety and notes

The maximum output current of 4WD Robot HAT is 4A. Please do not use it to drive high-power motors. If the power of the entire circuit is too high and the current exceeds 4A, the F1 fuse on the board will blow.

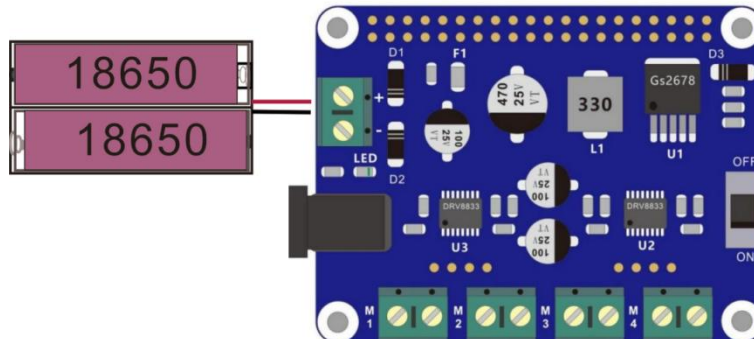
Before using 4WD Robot HAT, Raspberry Pi, and some other modules to build a circuit, be sure to check if there are any problems with the circuit connection, especially if there is a short circuit between 3.3V and GND, and if there is a short circuit between 5V and GND? If powered on in a short circuit state, it may cause damage to some components on 4WD Robot HAT and Raspberry Pi, thus preventing them from working properly.

3.2.6 Power supply

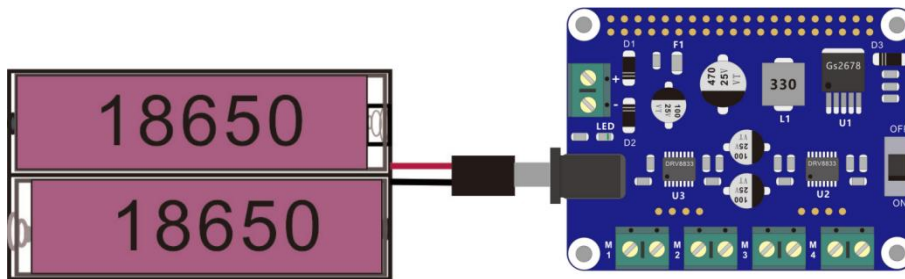
The 4WD Robot HAT board provides two power supply ports, one is a DC socket sub interface and the other is a 2-pin terminal. Any power supply port connected to a 7-12V battery or adapter can support the operation of the board, and can be matched with different types of batteries (such as AA batteries, 18650 batteries), battery packs, or adapters, making it convenient for users to use.

Taking the 18650 battery as an example, use a battery box to power the 4WD Robot HAT as shown in the following figure:

18650 battery box connected to 2-pin terminal for power supply



18650 battery box connected to DC socket for power supply



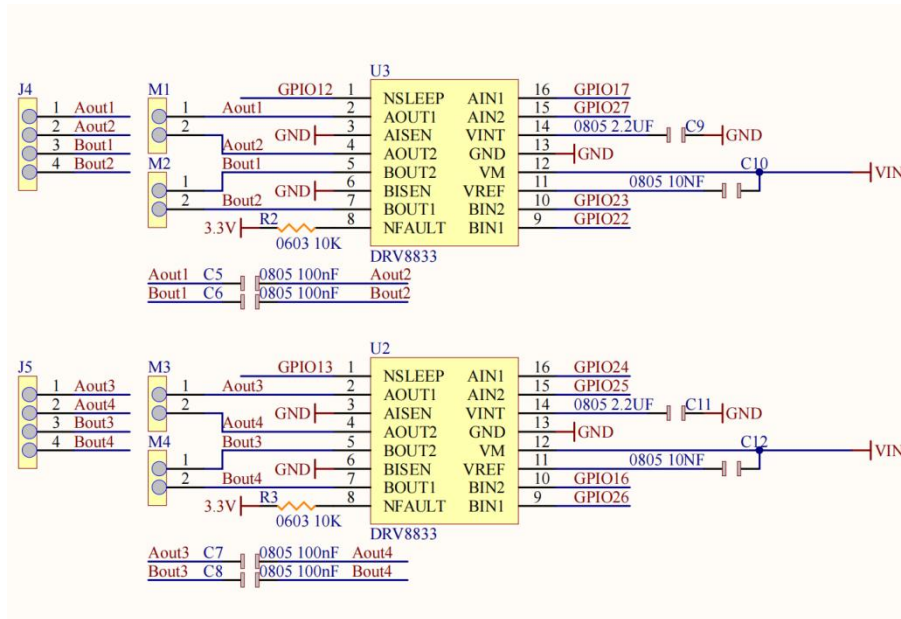
3.2.7 Pin Description of the 4WD Robot Hat

The 40 pins on the 4WD Robot HAT board correspond one-to-one to the 40 pins on the Raspberry Pi. Therefore, after inserting the 4WD Robot HAT onto the Raspberry Pi, the Raspberry Pi signal can be extended and used on the 4WD Robot HAT. It should be noted that Pi Power&4W HAT has already occupied some signal pins for driving the motor chip DRV8833

4W Robot HAT occupy the pins of Raspberry Pi as shown in the table below

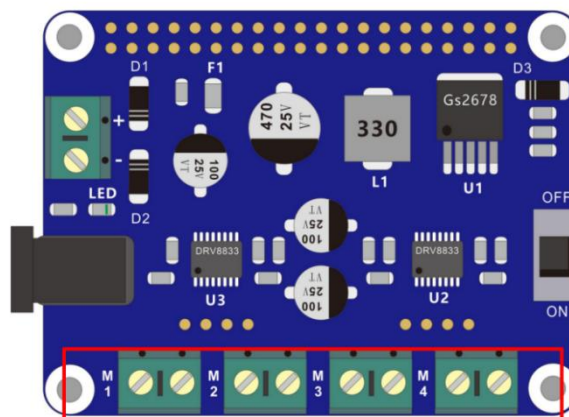
Pi Power & 4W HAT	Raspberry Pin
DRV8833#1 NSLEEP	GPIO12
DRV8833#1 AIN1	GPIO17
DRV8833#1 AIN2	GPIO27
DRV8833#1 BIN1	GPIO22
DRV8833#1 BIN2	GPIO23
DRV8833#2 NSLEEP	GPIO13
DRV8833#2 AIN1	GPIO24
DRV8833#2 AIN2	GPIO25
DRV8833#2 BIN1	GPIO26
DRV8833#2 BIN2	GPIO16

The schematic diagram of the Drv8833 circuit driven by the Raspberry Pi IO port signal is as follows



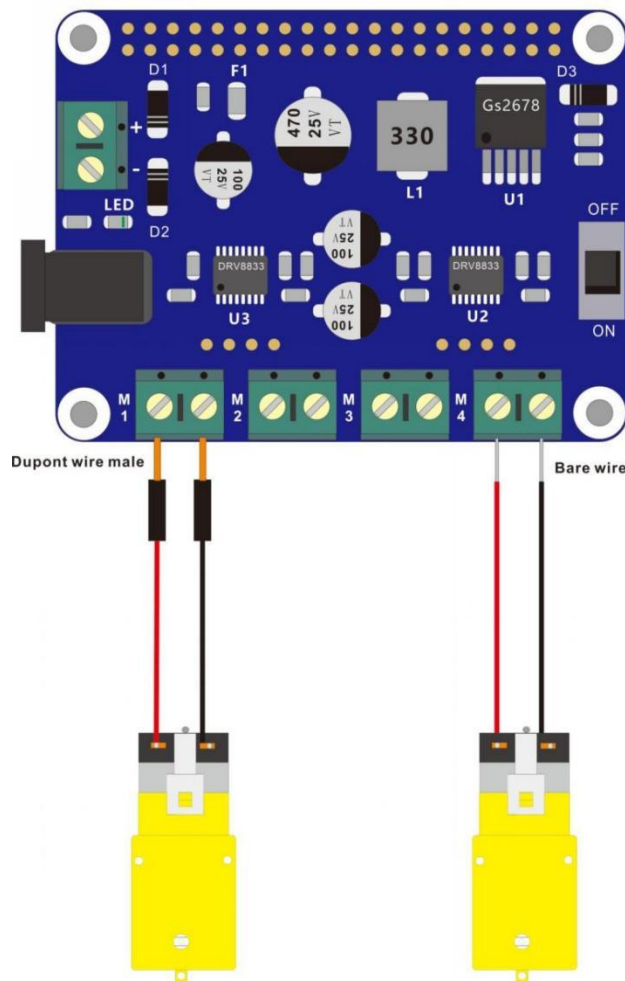
3.2.8 DC motor interfaces on the 4WD Robot Hat

The 4WD Robot HAT board supports driving 4 DC motors and is distributed with two types of motor interfaces, one is a 2-pin terminal interface and the other is a row pin interface. The 2-pin terminal interface is marked as follows:

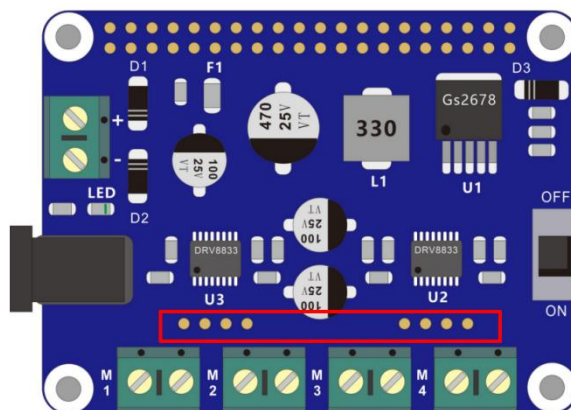


2-pin wiring terminal, suitable for DC motors with DuPont wire male or bare wire at the end of the wires

Taking TT motor as an example, as shown in the diagram below, the interface of 4WD Robot HAT connect motors which the wire end are DuPont wire male or bare wire .



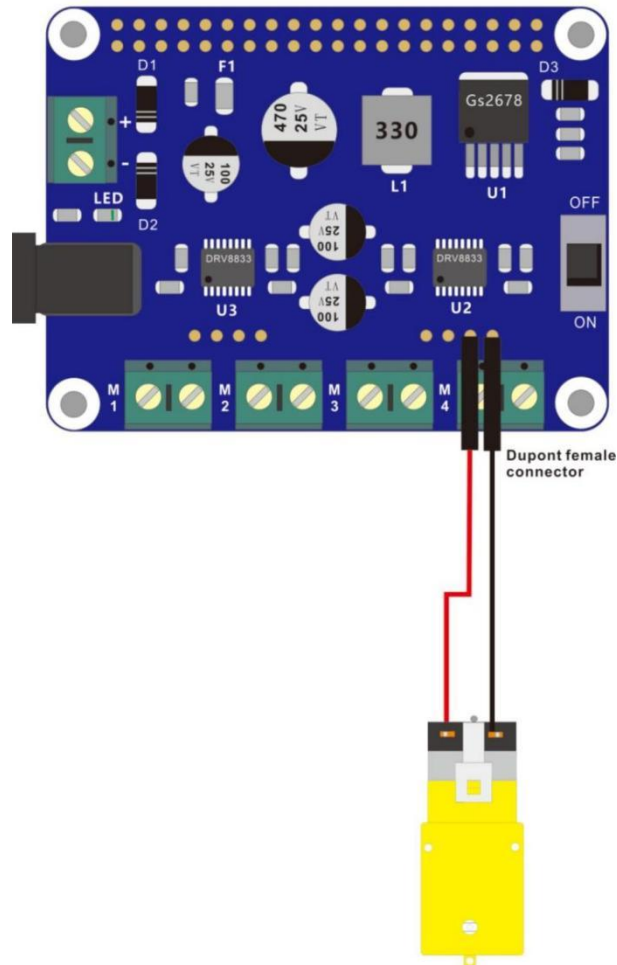
Another type of motor interface is row pin, suitable for connect the motors which end of the connection wire are DuPont female connectors.



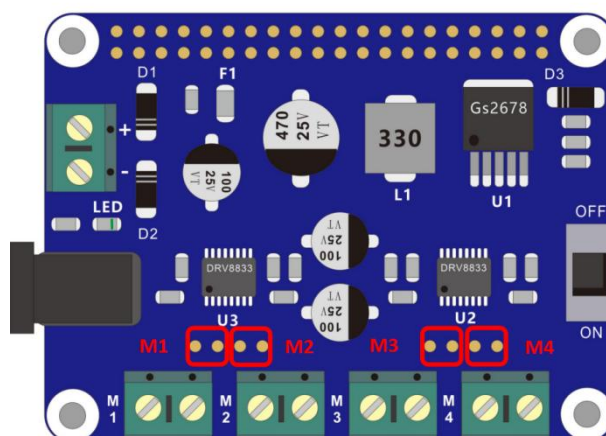
row pin interface for DC motor, suitable for DC motor with DuPont female connector at the end of the connecting wire

Taking TT motor as an example, as shown in the figure below, 4WD Robot HAT row pin interface

connect the TT motor which the end of connecting wire are DuPont female connector.



The motor serial number corresponding to the row pin interface is shown in the following figure



3.3 Knowledge of the Raspberry Pi

GPIO

GPIO: General Purpose Input/Output. Here we will introduce the specific function of the pins on the Raspberry Pi and how you can utilize them in all sorts of ways in your

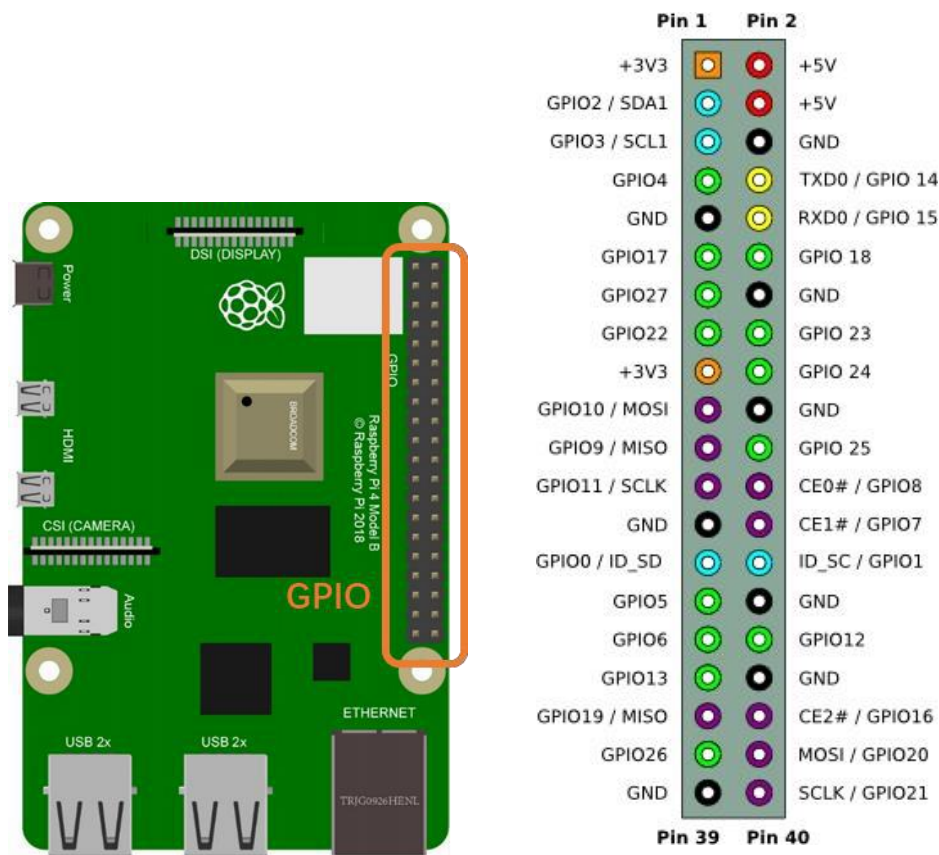
projects. Most RPi Module pins can be used as either an input or output, depending on your program and its functions.

When programming GPIO pins there are 3 different ways to reference them: GPIO Numbering, Physical Numbering and WiringPi GPIO Numbering.

BCM GPIO Numbering

The Raspberry Pi CPU uses Broadcom (BCM) processing chips BCM2835, BCM2836 or BCM2837. GPIO pin numbers are assigned by the processing chip manufacturer and are how the computer recognizes each pin. The pin numbers themselves do not make sense or have meaning as they are only a form of identification. Since their numeric values and physical locations have no specific order, there is no way to remember them so you will need to have a printed reference or a reference board that fits over the pins.

Each pin's functional assignment is defined in the image below:

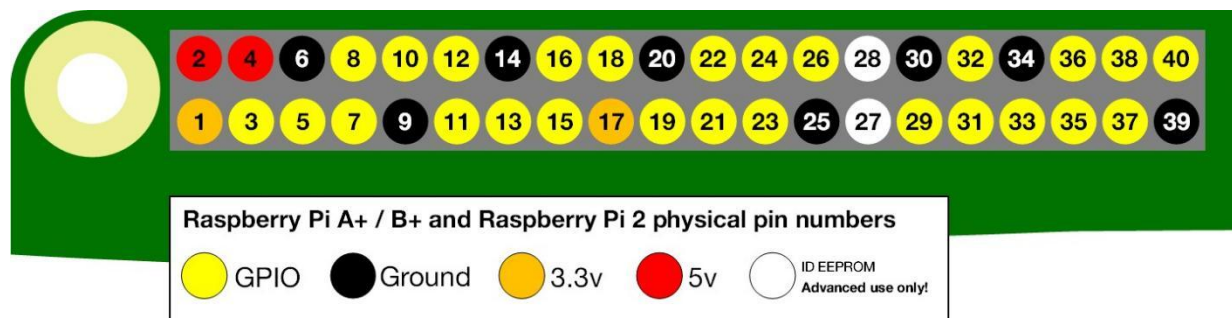


For more details about pin definition of GPIO, please refer to <http://pinout.xyz/>

PHYSICAL Numbering

Another way to refer to the pins is by simply counting across and down from pin 1 at

the top left (nearest to the SD card). This is 'Physical Numbering', as shown below:



WiringPi GPIO Numbering

Different from the previous two types of GPIO serial numbers, RPi GPIO serial number of the WiringPi are numbered according to the BCM chip use in RPi.

wiringPi Pin	BCM GPIO	Name	Header	Name	BCM GPIO	wiringPi Pin	
—	—	3.3v	1 2	5v	—	—	For Pi B
8	R1:0/R2:2	SDA	3 4	5v	—	—	
9	R1:1/R2:3	SCL	5 6	0v	—	—	
7	4	GPIO7	7 8	TxD	14	15	
—	—	0v	9 10	RxD	15	16	
0	17	GPIO0	11 12	GPIO1	18	1	
2	R1:21/R2:27	GPIO2	13 14	0v	—	—	
3	22	GPIO3	15 16	GPIO4	23	4	
—	—	3.3v	17 18	GPIO5	24	5	
12	10	MOSI	19 20	0v	—	—	
13	9	MISO	21 22	GPIO6	25	6	
14	11	SCLK	23 24	CE0	8	10	
—	—	0v	25 26	CE1	7	11	
30	0	SDA.0	27 28	SCL.0	1	31	
21	5	GPIO.21	29 30	0V	—	—	
22	6	GPIO.22	31 32	GPIO.26	12	26	
23	13	GPIO.23	33 34	0V	—	—	
24	19	GPIO.24	35 36	GPIO.27	16	27	
25	26	GPIO.25	37 38	GPIO.28	20	28	
—	—	0V	39 40	GPIO.29	21	29	
wiringPi Pin	BCM GPIO	Name	Header	Name	BCM GPIO	wiringPi Pin	

(For more details, please refer to <https://projects.drogon.net/raspberry-pi/wiringpi/pins/>)

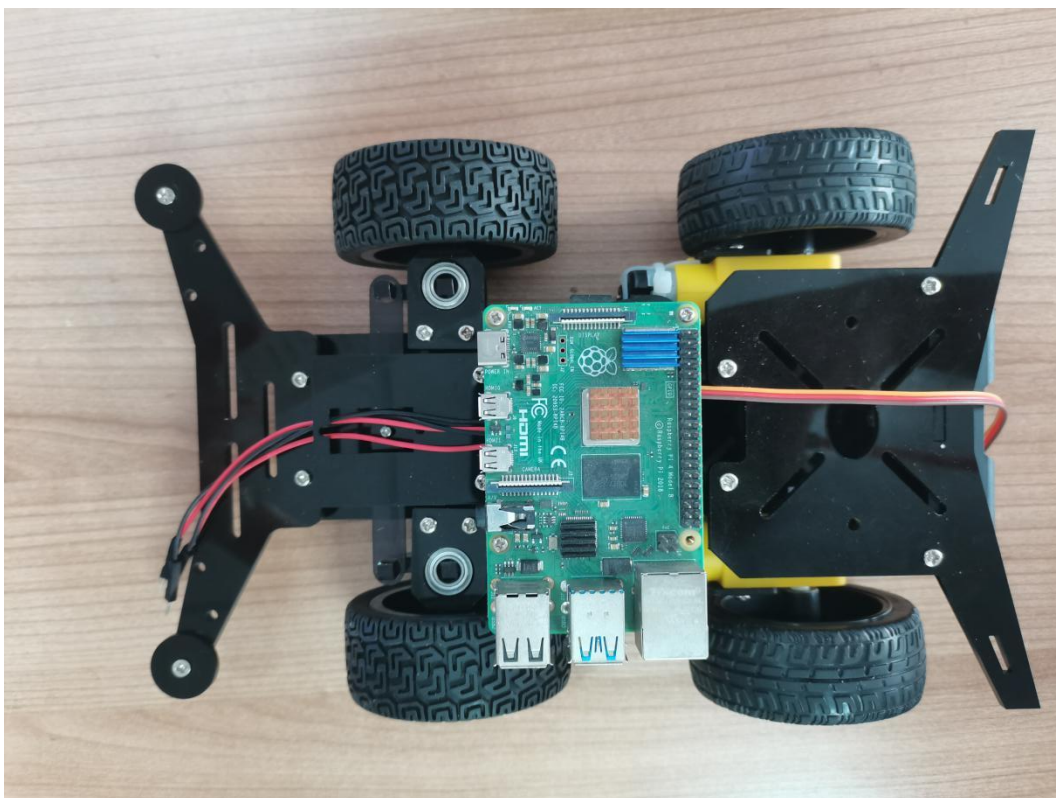
You can also use the following command to view their correlation.

```
gpio readall
```

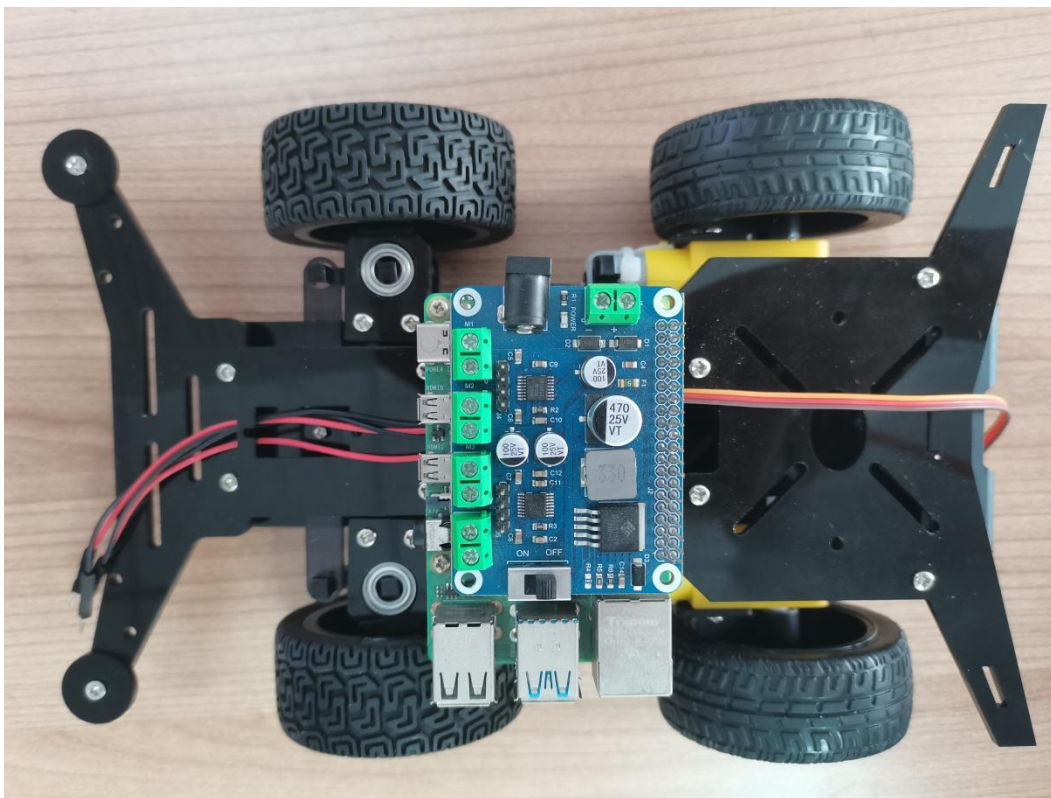
Pi 4B											
BCM	wPi	Name	Mode	V	Physical	V	Mode	Name	wPi	BCM	
		3.3v			1	2		5v			
2	8	SDA.1	ALT0	1	3	4		5v			
3	9	SCL.1	ALT0	1	5	6		0v			
4	7	GPIO. 7	IN	1	7	8	0	IN	TxD	15	14
		0v			9	10	1	IN	RxD	16	15
17	0	GPIO. 0	IN	0	11	12	0	IN	GPIO. 1	1	18
27	2	GPIO. 2	IN	0	13	14		0v			
22	3	GPIO. 3	IN	0	15	16	0	IN	GPIO. 4	4	23
		3.3v			17	18	0	IN	GPIO. 5	5	24
10	12	MOSI	IN	0	19	20		0v			
9	13	MISO	IN	0	21	22	0	IN	GPIO. 6	6	25
11	14	SCLK	IN	0	23	24	1	IN	CE0	10	8
		0v			25	26	1	IN	CE1	11	7
0	30	SDA.0	IN	1	27	28	1	IN	SCL.0	31	1
5	21	GPIO.21	IN	1	29	30		0v			
6	22	GPIO.22	IN	1	31	32	0	IN	GPIO.26	26	12
13	23	GPIO.23	IN	0	33	34		0v			
19	24	GPIO.24	IN	0	35	36	0	IN	GPIO.27	27	16
26	25	GPIO.25	IN	0	37	38	0	IN	GPIO.28	28	20
		0v			39	40	0	IN	GPIO.29	29	21
BCM	wPi	Name	Mode	V	Physical	V	Mode	Name	wPi	BCM	

4. Circuit Connection

First, turn off power of the 4WD Robot HAT,Fix the RaspebrryPi 4B onto the chassis using 4 pcs M2.5 * 8mm screws, as shown in the following figure:



Then insert the 4WD Robot Hat onto the Raspberry Pi 4B, paying attention to the

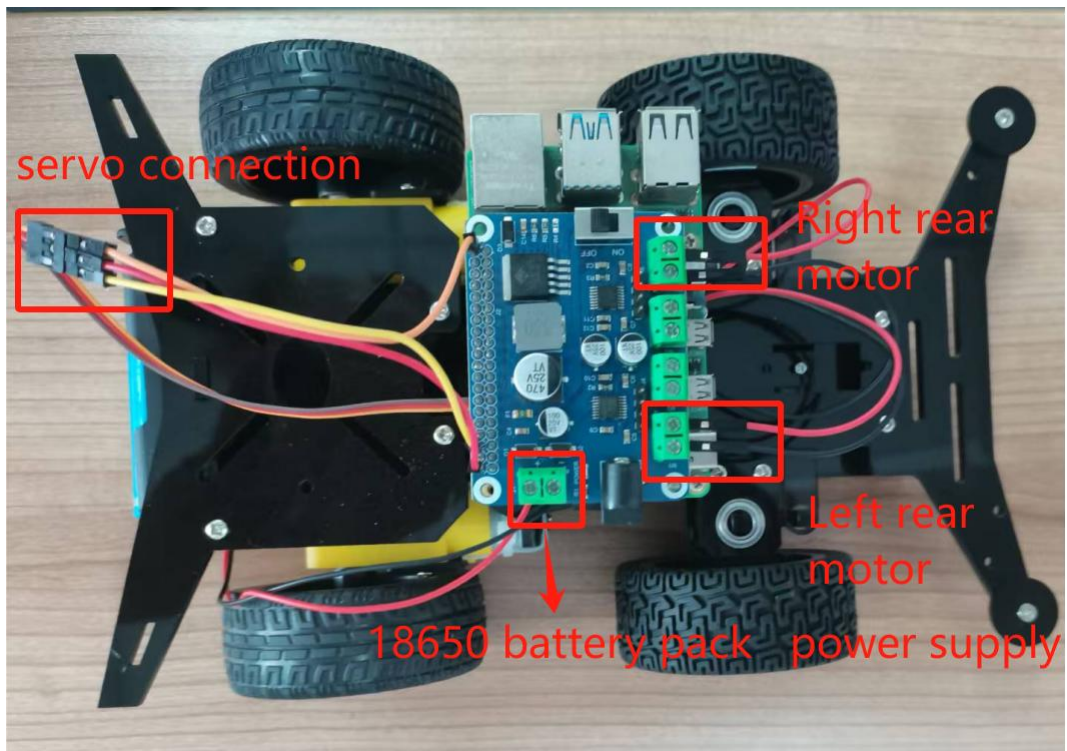
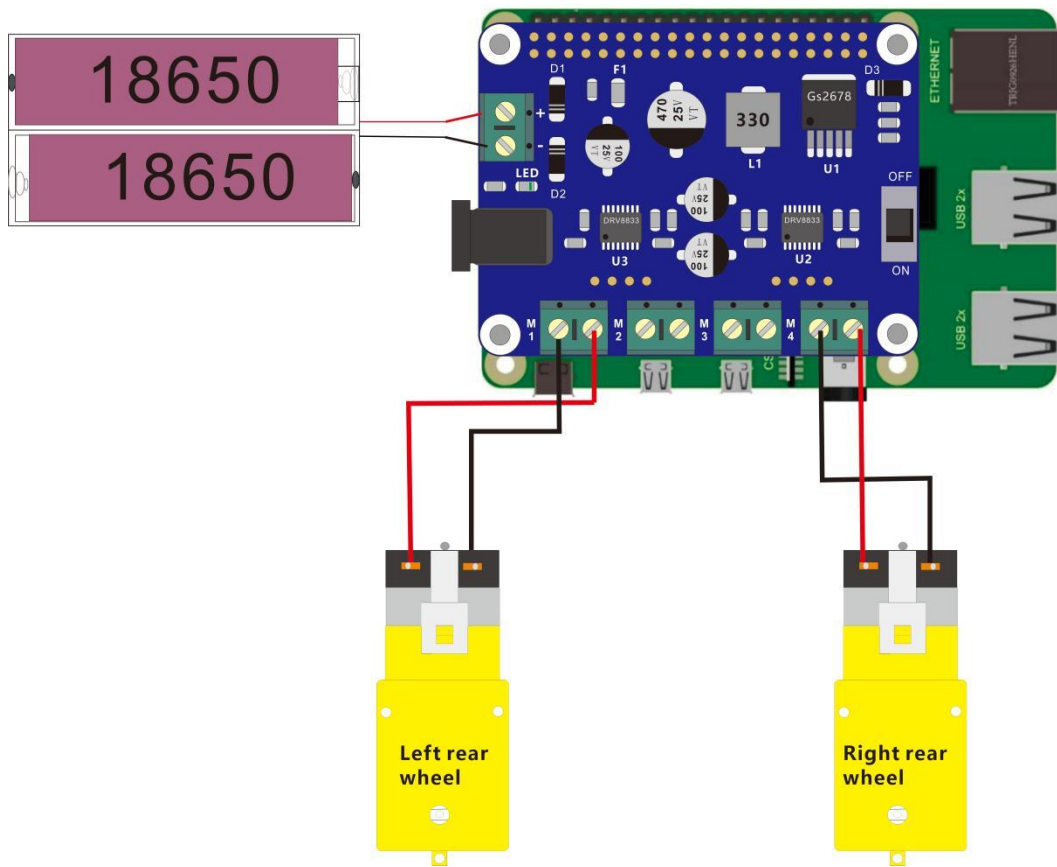


CAUTION: Avoid any possible short circuits (especially connecting 5V or GND, 3.3V and GND)!

17

Support : cokoino@outlook.com

4.1 Circuit connection diagram:



5. Download Code and Run

We recommend use Python code to achieve the function.

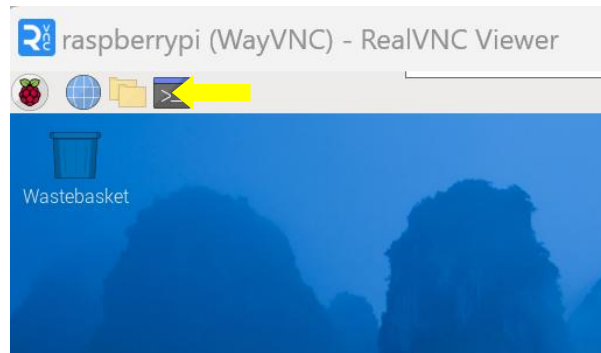
Please visit our GitHub resources at (<https://github.com/cokoino>) to download the latest available project code. We provide **Python** language code for this project .

This is the method for obtaining the code:

```
cd
git clone --depth 1 https://github.com/cokoino/CKK0020
```

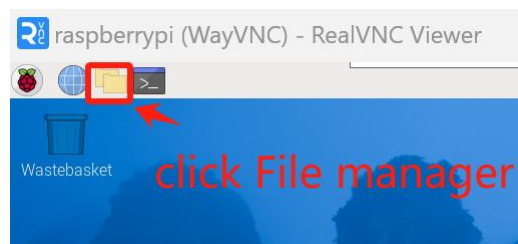
In the pi directory of the RPi terminal, enter the following command.

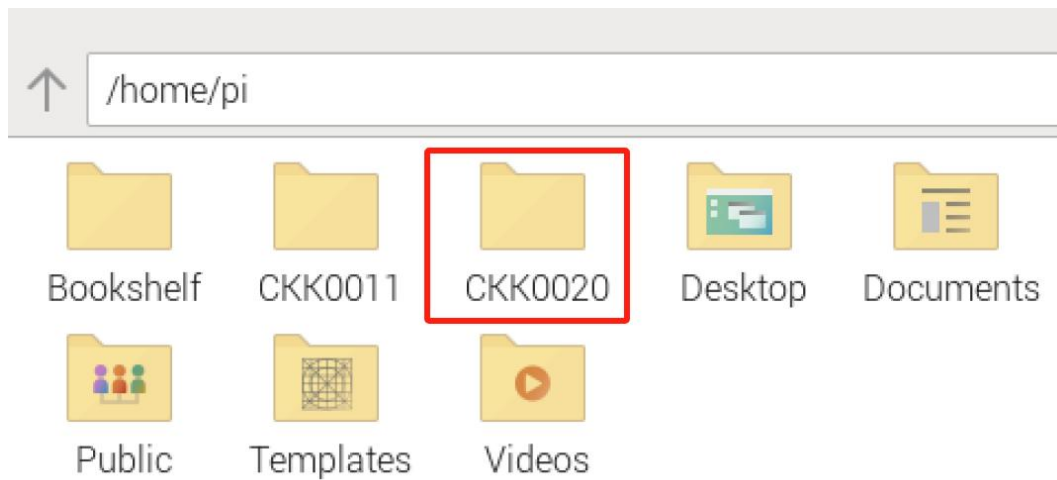
(There is no need for a password. If you get some errors, please check your commands.)



After the download is completed, a new folder "CKK0020" is generated, which contains all of the tutorials and required code.

Click File manager, you will find the folder "CKK0020"

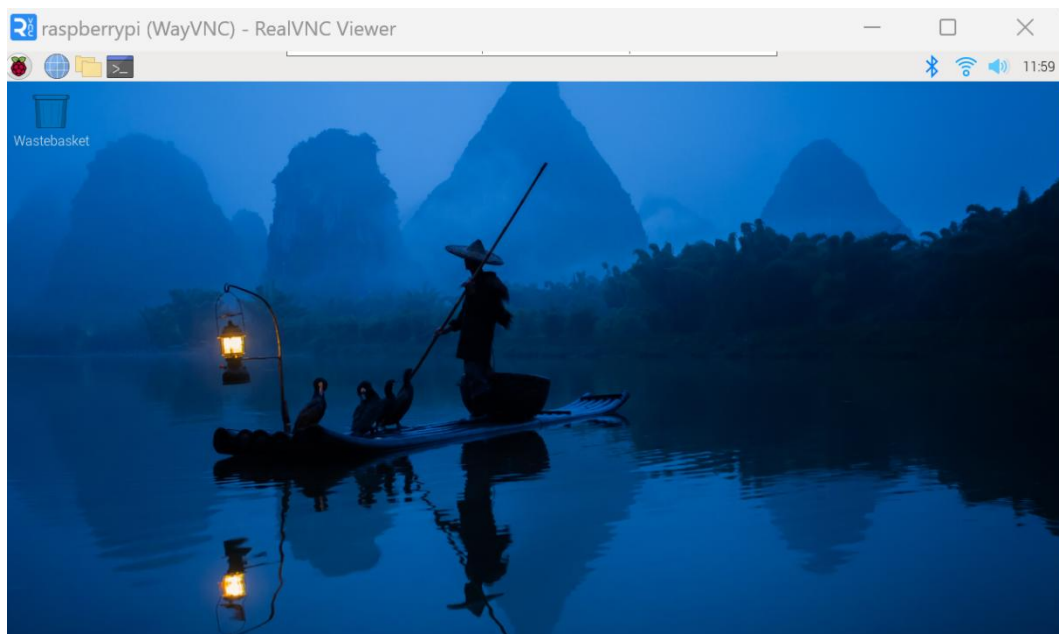




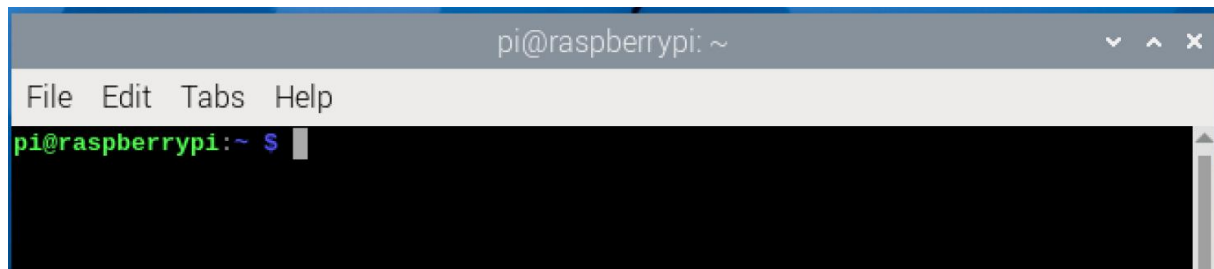
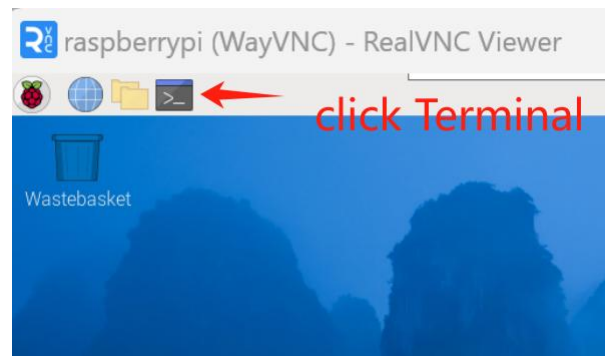
If you have no experience with Python, we suggest that you refer to this website for basic information and knowledge.

<https://python.swaroopch.com/basics.html>

Then, log in to Raspberry Pi by using VNC Viewer and operated proper setting. (For those who have no impression or are unclear about the operation, please refer to the PDF document "[Lesson 1 Installing and Configuring Raspberry Pi System](#)").



Click “Terminal”, following interface appears.



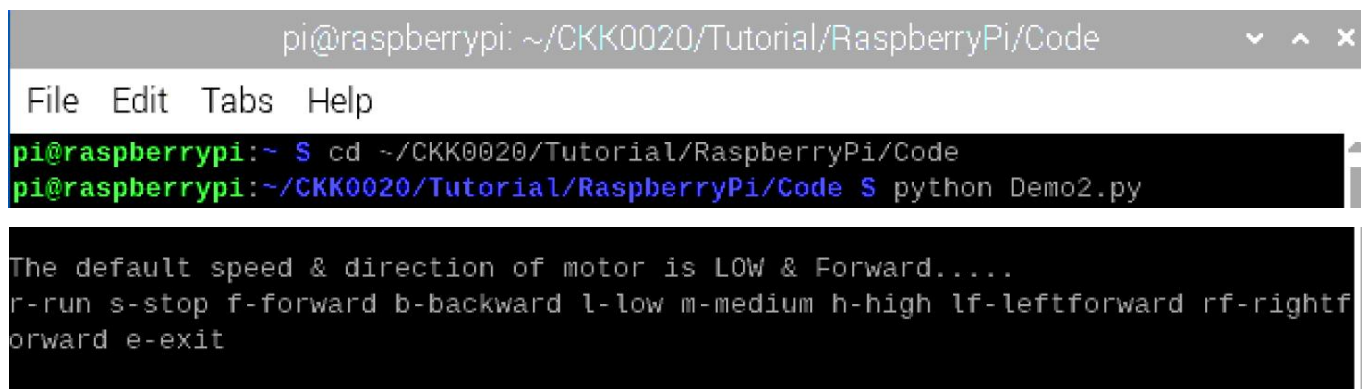
Use cd command to enter Demo1 directory of Python code.

```
cd ~/CKK0020/Tutorial/RaspberryPi/Code
```

Use python command to execute python code Demo2.py.

```
python Demo2.py
```

Then the interface displays “The default direction of motor is LOW & Forward.....r-run s-stop f-forward b-backward l-low m-medium h-high lf-lefforward rf-rightforward e-exit”



Finally, Enter the command according to the prompt information and press enter to observe whether the car will perform the corresponding action. For example, after entering "f" and press enter, the car starts to move forward with low speed;After entering "m"and press enter, the car moves with midium speed;after entering "lf" and press enter, the car starts to move leftforward;after entering "rf" and press enter, the car starts to move rightforward;After entering "s"and press enter, the car stops moving.

At the same time, the remote desktop interface has corresponding information display.

```
pi@raspberrypi: ~/CKK0020/Tutorial/RaspberryPi/Code
File Edit Tabs Help
le warnings.
GPIO.setup(BN22,GPIO.OUT)
/home/pi/CKK0020/Tutorial/RaspberryPi/Code/Demo2.py:29: RuntimeWarning: This cha
channel is already in use, continuing anyway. Use GPIO.setwarnings(False) to disab
le warnings.
GPIO.setup(servo_pin, GPIO.OUT)

The default speed & direction of motor is LOW & Forward.....
r-run s-stop f-forward b-backward l-low m-medium h-high lf-leftforward rf-rightf
orward e-exit

f
forward
m
medium
lf
leftforward
rf
rightforward
s
stop
```

You can enter the“e”command or press “Ctrl+C” to end the program.

About RPi.GPIO:

RPi.GPIO	This is a Python module to control the GPIO on a Raspberry Pi. It includes basic output function and input function of GPIO, and functions used to generate PWM.
GPIO.setmode(mode)	Sets the mode for pin serial number of GPIO. mode=GPIO.BOARD, which represents the GPIO pin serial number based on physical location of RPi. mode=GPIO.BCM, which represents the pin serial number based on CPU of BCM chip.
GPIO.setup(pin,mode)	Sets pin to input mode or output mode, “pin” for the GPIO pin, “mode” for INPUT or OUTPUT.
GPIO.output(pin,mode)	Sets pin to output mode, “pin” for the GPIO pin, “mode” for HIGH (high level) or LOW (low level).

For more functions related to RPi.GPIO, please refer to:

<https://sourceforge.net/p/raspberry-gpio-python/wiki/Examples/>

“import time” time is a module of python.

<https://docs.python.org/2/library/time.html?highlight=time%20time#module-time>

In subfunction setup(), GPIO.setmode (GPIO.BOARD) is used to set the serial number for GPIO based on physical location of the pin.

GPIO Numbering Relationship

WingPi	BCM(Extension)	Physical		BCM(Extension)	WingPi
3.3V	3.3V	1	2	5V	5V
8	GPIO2/SDA1	3	4	5V	5V
9	GPIO3/SCL1	5	6	GND	GND
7	GPIO4	7	8	GPIO14/TXD0	15
GND	GND	9	10	GPIO15/RXD0	16
0	GPIO17	11	12	GPIO18	1
2	GPIO27	13	14	GND	GND
3	GPIO22	15	16	GPIO23	4
3.3V	3.3V	17	18	GPIO24	5
12	GPIO10/MOSI	19	20	GND	GND
13	GPIO9/MOIS	21	22	GPIO25	6
14	GPIO11/SCLK	23	24	GPIO8/CE0	10
GND	GND	25	26	GPIO7/CE1	11
30	GPIO0/SDA0	27	28	GPIO1/SCL0	31
21	GPIO5	29	30	GND	GND
22	GPIO6	31	32	GPIO12	26
23	GPIO13	33	34	GND	GND
24	GPIO19	35	36	GPIO16	27
25	GPIO26	37	38	GPIO20	28
GND	GND	39	40	GPIO21	29

6. Trouble shooting

Question 1: If the car does not move according to the code instructions, such as the command being forward, but the actual action of the car is backward or other actions.

Answer: Please check if the motors on the rear wheels of the car are connected to the corresponding positions of the Cokoino 4WD Robot Hat Shield. If not, the car will not perform the corresponding actions according to the instructions.

Question2: Using RealVNC Viewer as a remote desktop connection does not connect to Raspberry Pi, and Raspberry Pi does not display when connected to the monitor.

Answer: Please check the battery level that supplies power to the Cokoino 4WD Robot Hat Shield. When the battery voltage is below 6.8V, the Raspberry Pi cannot be powered on. Please fully charge the battery before use.

7. Any questions and suggestions are welcome

THANK YOU for participating in this Demo2 experience!

If you find any errors, omissions or you have suggestions and/or questions about this lesson, please feel free to contact us:

cokoino@outlook.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

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